MODULAR KOSZUL DUALITY: SECTIONS 9–14

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ABSTRACT. Sections 9–14 of the programme summary instantiate the modular Koszul machinery on the standard landscape and trace its consequences across the holographic, analytic, and arithmetic frontiers. Section 9 catalogues the seven chirally Koszul families; Section 10 transports the bar–cobar machine into three dimensions via the Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$; Sections 11–12 package the closed-sector/boundary/line algebraic triangle into a single holographic datum controlled by a universal Maurer–Cartan element, with physical BBL language only after OCA; Section 13 sketches the completion and analytic sewing programmes; Section 14 surveys the arithmetic of the shadow obstruction tower and the residual open frontiers. The closing scope is the non-logarithmic C_2-cofinite standard landscape plus irrational cosets; logarithmic $W(p)$ tempering, the exact closed form of β_N at $N \geq 4$, the canonical super-complementarity Verdier pairing, and chain-level class M on the original complex are the four residual frontiers.

1. The standard landscape

The bar construction, the genus tower, the Swiss-cheese structure, and the collision residue are not abstractions awaiting examples: they were *found* by computing examples. The standard landscape is the testing ground where the theory proves itself.

1.1. **Census.** Seven families span the landscape. Each is chirally Koszul. The invariants computed in the monograph are recorded in Table 1.

TABLE 1. The standard landscape: seven families and their bar-complex invariants

Algebra $\mathcal{A}$	$c(\mathcal{A})$	$\kappa(\mathcal{A})$	$r_{\max}$	**Class**	$r(z)$	$\mathcal{A}^!$
$\mathcal{H}_k$	1	k	2	**G**	k/z	$\mathrm{Sym}^{\mathrm{ch}}(V^*)$
$\widehat{\mathfrak{g}}_k$	$\dfrac{kd}{k+h^\vee}$	$\dfrac{d(k+h^\vee)}{2h^\vee}$	3	**L**	$\Omega/((k+h^\vee)z)$	$\widehat{\mathfrak{g}}_{-k-2h^\vee}$
$\beta\gamma_\lambda$	$1-3(2\lambda-1)^2$	$c/2$	4	**C**	const.	bc_λ
Vir_c	c	$c/2$	∞	**M**	$\frac{c/2}{z^3}+\frac{2T}{z}$	Vir_{26-c}
$\mathcal{W}_3^k(\mathfrak{sl}_3)$	$c(k)$	$5c/6$	∞	**M**	$r_{\mathcal{W}_3}(z)$	$\mathcal{W}_3^{k'}$
$\mathcal{W}_N^k(\mathfrak{sl}_N)$	$c(k)$	$c\cdot(H_N-1)$	∞	**M**	$r_{\mathcal{W}_N}(z)$	$\mathcal{W}_N^{k'}$
V_Λ (lattice)	$\mathrm{rank}(\Lambda)$	$\mathrm{rank}(\Lambda)$	2	**G**	rank/z	V_{Λ^*}

Here $d = \dim\mathfrak{g}$, $h^\vee$ is the dual Coxeter number, $t = k + h^\vee$, $k' = -k - 2h^\vee$, $H_N = \sum_{j=1}^N 1/j$ is the N-th harmonic number, and $\Omega = \sum_a J^a \otimes J_a$ is the Casimir tensor.

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The table has two readings. Read horizontally, each row is a complete portrait: central charge, modular characteristic, shadow depth, collision residue, Koszul dual. Read vertically, the columns expose the structural regularities that no single example can reveal. The modular characteristic κ is always a rational function of the level; the complementarity sum $\kappa + \kappa^!$ vanishes for free-field and Kac–Moody algebras, equals 13 for Virasoro, and is $(c + c')(H_N - 1)$ for $\mathcal{W}_N$. The collision kernel $K^{\mathrm{coll}}_{\mathcal{A}}(z)$ is scalar for classes **G** and **C**, matrix-valued for classes **L** and **M**.

1.2. **The DS reduction chain.** Drinfeld–Sokolov reduction links the families. For $\mathfrak{g} = \mathfrak{sl}_N$, the principal DS reduction

$$V_k(\mathfrak{sl}_N) \xrightarrow{\mathrm{DS}} \mathcal{W}^k_N(\mathfrak{sl}_N)$$

transforms the shadow obstruction tower in a controlled way:

(i) *Depth increase.* The affine algebra $V_k(\mathfrak{sl}_N)$ is class **L** ($r_{\max} = 3$); the $\mathcal{W}$-algebra $\mathcal{W}^k_N$ is class **M** ($r_{\max} = \infty$). DS reduction creates the infinite shadow tower from finite data.

(ii) *Quartic creation from zero.* The quartic shadow S_4 vanishes for all affine algebras (the Jacobi identity kills the obstruction); after DS reduction, $S_4(\mathcal{W}^k_N) \neq 0$ for generic k. For Virasoro: $S_4(\mathrm{Vir}_c) = 10/[c(5c + 22)]$.

(iii) *Central charge is additive; κ is not.* The BRST ghost system contributes $c_{\mathrm{ghost}} = N(N-1)$ to the central charge but a k-dependent correction to κ. The commutation failure of DS reduction with the shadow map is the source of all non-formality in the $\mathcal{W}$-algebra landscape.

The chain $\mathcal{H} \to \widehat{\mathfrak{g}}_k \to \mathcal{W}^k_N$ is a cascade of increasing shadow complexity: **G** $\to$ **L** $\to$ **M**, with each reduction introducing new structure that the previous level could not see.

1.3. **The multi-weight correction.** For algebras with generators of distinct conformal weights, the genus expansion receives a correction. The $\mathcal{W}_3$ algebra has generators T (weight 2) and W (weight 3); at genus 2 the cross-channel graph sum contributes

$$\delta F_2(\mathcal{W}_3) = \frac{c + 204}{16c} > 0 \qquad \text{for all } c > 0\,.$$

The correction arises from boundary graphs of $\overline{\mathcal{M}}_{2,0}$ whose edges carry mixed propagator channels (weight-2 and weight-3 generators sewed through different Hodge data at the vertex level). The propagator itself is always weight-1 at the edge level ($d \log E$ has conformal weight 1); the mixing enters through the vertex structure constants.

For uniform-weight algebras (all generators of the same conformal weight), $\delta F^{\mathrm{cross}}_g = 0$ at all genera and the scalar universality $F_g = \kappa \cdot \lambda^{\mathrm{FP}}_g$ holds without correction. For multi-weight algebras, the full genus expansion is $F_g = \kappa \cdot \lambda^{\mathrm{FP}}_g + \delta F^{\mathrm{cross}}_g$, with $\delta F^{\mathrm{cross}}_g$ computable from the mixed-channel graph sum.

2. The MC programme

The MC programme converts the geometric structure of the bar complex into five concrete mathematical problems, each asking whether a structural property of the universal MC element $\Theta_{\mathcal{A}}$ persists under a specific operation: PBW concentration, bar-intrinsic existence, categorical generation, inverse-limit completion, and analytic sewing. MC1 through MC4 are proved; MC5 is partially proved, with the analytic HS-sewing package established at all genera and the genuswise BV/BRST/bar identification conjectural.

2.1. **MC1: PBW concentration.** Free strong generation makes the PBW filtration on $B_X(\mathcal{A})$ exhaustive and identifies its associated graded bar coalgebra. Collapse and detection require the separate package $H^{\mathrm{det}}_{\mathrm{PBW}}$. For a chosen presentation $\mathcal{A} = T_X(V)/(R)$, this package promotes the canonical comparison

$$q_{\mathcal{A}} : \mathcal{A}^{\mathrm{i}} = C_X(s^{-1}V, s^{-2}R) \longrightarrow B_X(\mathcal{A})$$

to a quasi-isomorphism. Consequently $H^\bullet(q_{\mathcal{A}})$ identifies presentation-coalgebra cohomology with bar cohomology.

Feigin–Frenkel free generation supplies the PBW input for the universal $\mathcal{W}$-algebra $\mathcal{W}^k(\mathfrak{g})$. Each asserted collapse also carries $H^{\mathrm{det}}_{\mathrm{PBW}}$. For simple quotients $L_k(\mathfrak{g})$ at admissible levels, the rank-one $\mathfrak{sl}_2$ lane uses the quotient-bar spectral sequence and convergence package; higher rank carries the Cartan obstruction described in the main text.

2.2. **MC2: bar-intrinsic existence.** The universal MC element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$ exists for all modular Koszul chiral algebras, with no restriction to simple Lie symmetry. The proof is a tautology at the right level of abstraction: $D^2_{\mathcal{A}} = 0$ implies $\Theta_{\mathcal{A}}$ satisfies the MC equation (**??**). The algebraic-family rigidity theorem extends the scalar identification $\Theta^{\min}_{\mathcal{A}} = \eta \otimes \Gamma_{\mathcal{A}}$ to all algebraic families with rational OPE coefficients, covering the entire standard Lie-theoretic landscape at all non-critical levels.

2.3. **MC3: categorical generation.** For every simple Lie algebra $\mathfrak{g}$, the Drinfeld–Kazhdan category $\mathrm{DK}(\mathfrak{g})$ is thickly generated by evaluation modules. The proof combines the chromatic filtration on $\mathrm{Rep}(Y(\mathfrak{g}))$ with the categorical Clebsch–Gordan closure via multiplicity-free ℓ-weights (Chari–Moura), the Francis–Gaitsgory pro-nilpotent completion, and Drinfeld–Kazhdan compactness. The type A proof uses the minuscule hypothesis; the all-types generalization replaces it with the strictly weaker multiplicity-free ℓ-weight property, which holds for all simple types.

2.4. **MC4: inverse-limit completion.** The bar-cobar adjunction extends to inverse limits via the strong completion-tower theorem: the strong filtration axiom $\mu_r(F^{i_1}, \ldots, F^{i_r}) \subset F^{i_1+\cdots+i_r}$ forces a degree cutoff that makes continuity and the Mittag-Leffler condition automatic. The completion closure $\mathrm{CompCl}(F_{\mathrm{ft}})$ carries a quasi-inverse bar-cobar homotopy equivalence, stable under MC twisting.

The programme splits: $\mathrm{MC4}^+$ (positive towers: $\mathcal{W}_{1+\infty}$, affine Yangians) is solved by weight stabilization. $\mathrm{MC4}^0$ (resonant towers: Virasoro, non-quadratic $\mathcal{W}_N$) is reduced to a finite resonance problem by the resonance-filtered bar-cobar theorem.

2.5. **MC5: analytic sewing.** The sewing envelope $\mathcal{A}^{\mathrm{sew}}$, the Hausdorff completion of $\mathcal{A}$ for the all-amplitude seminorm topology, carries the analytic data needed for genus-g partition functions. The HS-sewing criterion provides a sufficient condition: polynomial OPE growth and subexponential sector growth imply convergence at all genera. Every standard family satisfies this condition.

For Heisenberg, the sewing theorem is explicit: the one-particle Bergman reduction writes the genus-g partition function as a Fredholm determinant. For general modular Koszul algebras, the HS-sewing criterion gives convergence without an explicit closed form. The BV/BRST identification of sewing amplitudes with bar complex data at higher genus remains conjectural.

3. Arithmetic

The shadow obstruction tower produces arithmetic invariants from vertex-algebraic data. These invariants connect the bar complex to number theory through three structures: a Dirichlet series, a categorical zeta function, and a depth decomposition.

3.1. **The shadow Eisenstein theorem.** The genus-1 amplitude of a modular Koszul chiral algebra $\mathcal{A}$ is the quasi-modular Eisenstein series $\kappa(\mathcal{A}) \cdot E_2^*(\tau)$, where $E_2^*(\tau) = E_2(\tau) - 3/(\pi \operatorname{Im} \tau)$ is the completed weight-2 Eisenstein series. The Fourier coefficients $\sigma_1(n)$ of E_2^* produce a Dirichlet series:

$$D_2(\mathcal{A}, s) \;:=\; -24\kappa(\mathcal{A}) \cdot \zeta(s) \cdot \zeta(s-1)\,,$$

an Eisenstein L-function with poles at $s = 1$ (from $\zeta(s)$) and $s = 2$ (from $\zeta(s-1)$). The identity follows from the σ_1 Dirichlet convolution applied to the Fourier coefficients of $\kappa \cdot E_2^*(\tau)$.

The genus-1 propagator is E_2^*, not a holomorphic modular form: E_2^* is quasi-modular of weight 2 and depth 1, transforming with an additive anomaly $3\tau/(\pi i)$ under $\mathrm{SL}(2, \mathbb{Z})$. Products of E_2^* live in the ring of quasi-modular forms $\mathbb{C}[E_2^*, E_4, E_6]$, not in the ring of holomorphic modular forms $\mathbb{C}[E_4, E_6]$. This distinction is load-bearing: holomorphic dimension formulas (e.g., $\dim S_k = 0$ for $k < 12$) do not apply to quasi-modular objects.

3.2. **The categorical zeta function.** The dimension-counting zeta function of $\mathfrak{sl}_2$-representations is

$$\zeta^{\mathrm{DK}}_{\mathfrak{sl}_2}(s) \;:=\; \sum_{n \geq 1} (\dim V_n)^{-s} \;=\; \sum_{n \geq 1} (n+1)^{-s} \;=\; \zeta(s) - 1\,.$$

The Riemann zeta function (shifted by 1) counts $\mathfrak{sl}_2$-representations weighted by inverse dimension. The link to the shadow–DK bridge is that the categorical zeta encodes the multiplicity structure of the evaluation modules whose thick generation proves MC3.

For $\mathfrak{sl}_N$ with $N \geq 3$, $\zeta^{\mathrm{DK}}_{\mathfrak{sl}_N}(s) := \sum_\lambda (\dim V_\lambda)^{-s}$ is not multiplicative: the Weyl dimension formula is a product of linear forms, but the sum over the dominant chamber has no Euler product. The $N = 2$ coincidence with $\zeta(s)$ is a low-rank accident.

3.3. **Depth decomposition.** The total shadow depth decomposes:

$$d(\mathcal{A}) \;=\; 1 + d_{\mathrm{arith}}(\mathcal{A}) + d_{\mathrm{alg}}(\mathcal{A})\,,$$

where d_{arith} counts arithmetic contributions (cusp forms in the automorphic decomposition of the shadow generating function) and d_{alg} counts algebraic contributions (non-formality of the L_∞ structure). The algebraic depth $d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$ recovers the G/L/C/M classification: $d_{\mathrm{alg}} = 0$ for class **G**, $d_{\mathrm{alg}} = 1$ for class **L**, $d_{\mathrm{alg}} = 2$ for class **C**, $d_{\mathrm{alg}} = \infty$ for class **M**. Shadow depths ≥ 5 are purely arithmetic: they arise from cusp forms, beginning with the weight-12 cusp form Δ_{12} at the genus-2 level.

The decomposition separates the algebraic content (governed by the MC equation on the cyclic deformation complex) from the arithmetic content (governed by automorphic forms on moduli of curves). The two are independent: a class-**G** algebra (algebraically trivial) can have nontrivial arithmetic depth, and a class-**M** algebra (algebraically infinite) can have vanishing arithmetic depth at low genus.

4. Frontier directions

Three leaps organize the frontier. The first extends the modular Koszul programme from chiral algebras to holographic systems. The second extends it from formal algebraic data to analytic sewing amplitudes. The third extends it from 2-dimensional geometry to 3-dimensional topology.

4.1. **First leap: holographic modular Koszul data.** Every holomorphic-topological system T in the sense of Costello–Li should be compared with a *holographic modular Koszul datum* only after the typed open–closed and line-comparison maps are supplied:

$$\mathcal{H}(T) \;=\; (\mathcal{A},\, \mathcal{A}^{\mathrm{i}},\, \mathcal{A}^{!},\, Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}),\, r(z),\, \Theta_{\mathcal{A}},\, \nabla^{\mathrm{hol}})\,,$$

The boundary presentation $\mathcal{A} = T_X(V)/(R)$ supplies $\mathcal{A}^{\mathrm{i}} = C_X(s^{-1}V, s^{-2}R)$ and the comparison $q_{\mathcal{A}} \colon \mathcal{A}^{\mathrm{i}} \to B_X(\mathcal{A})$. Verdier duality forms $K_X(\mathcal{A}) = \mathbb{D}_{\mathrm{Ran}} B_X(\mathcal{A})$, and a chosen Koszul-pair map

$\nu_{\mathcal{A}}\colon K_X(\mathcal{A}) \to \mathcal{A}^!$ identifies the strict partner on the finite-type or continuously completed comparison surface. The remaining entries are the algebraic closed-sector object $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$, the collision residue, the universal MC element, and the holographic shadow connection. The line category is $\mathcal{A}^!$-modules only on the strict line-comparison surface; a physical bulk requires the open–closed comparison map. The collision residue is the genus-0 binary shadow of $\Theta_{\mathcal{A}}$: $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$.

This is the Platonic holographic programme. Its five theorem targets are: boundary-defect realization (the boundary algebra is the algebra of boundary conditions of the 3d theory), Yangian-shadow identification (the Yangian controls line operators), sphere reconstruction (genus-0 amplitudes recover the boundary OPE), quartic resonance obstruction (the quartic shadow controls the deformation quantization), and singular-fibre descent (the critical-level limit produces the Feigin–Frenkel centre).

Two other directions in the first leap: the factorization-envelope technology of Nishinaka and Vicedo, which gives a functorial pipeline from Lie conformal algebras to vertex algebras; and the non-principal $\mathcal{W}$-algebra corridor, where hook-type nilpotent orbits in type A provide the first proved non-principal Koszul dualities.

4.2. **Second leap: analytic sewing.** The algebraic bar construction produces formal power series in the sewing parameters. The analytic sewing programme asks: when do these series converge?

The answer is the sewing envelope $\mathcal{A}^{\mathrm{sew}}$, the Hausdorff completion for the all-amplitude seminorm topology. The HS-sewing condition (polynomial OPE growth and subexponential sector growth) is satisfied by the entire standard landscape. The Heisenberg sewing theorem gives an explicit Fredholm determinant. The lattice sewing envelope is constructed; the analytic realization criterion (a sufficient condition for passage from formal to analytic) is conjectural.

At genus $g \geq 1$ the curvature (??) forces passage from ordinary to coderived categories: the bar complex is no longer flat, and the correct homological framework is the coderived category of curved coalgebras. The analytic programme at higher genus requires the coderived analytic shadow $Q^{\mathrm{an}}_g(\mathcal{A})$.

4.3. **Third leap: three-dimensional topology.** The ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ is an E_1 chiral coassociative coalgebra over $(\mathrm{ChirAss})^!$: the differential encodes holomorphic factorization on $\mathbb{C}$, the deconcatenation coproduct encodes topological factorization on $\mathbb{R}$. The two-coloured Swiss-cheese operad $\mathrm{SC}^{\mathrm{ch,top}}$ of a 3-dimensional holomorphic-topological theory emerges on the chiral derived center pair $(C^{\bullet}_{\mathrm{ch}}(\mathcal{A},\mathcal{A}),\ \mathcal{A})$, not on $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ itself.

The third leap is the passage from the algebraic Swiss-cheese structure (proved) to the geometric realization on 3-manifolds of the form $\Sigma_g \times [0,1]$. The $\mathbb{C}$-direction carries the chiral algebra; the $\mathbb{R}$-direction carries the Yangian / quantum group. The annulus trace $\Delta_{\mathrm{ns}}(\mathrm{Tr}_{\mathcal{A}}) = \kappa \cdot \lambda_1$ provides the first open-to-closed map at genus 1; the full genus tower progressively restores bidirectionality between open and closed sectors.

The deepest open problem in this direction is CY-A at $d = 3$ (Section 5).

5. The organizing ambient problem

Everything in Sections ??–4 rests on two-dimensional geometry: curves, their moduli spaces, and the factorization structure of colliding points on a curve. The three-dimensional extension is now proved.

Theorem 5.1 (CY-A at $d = 3$). *The Calabi–Yau/A_∞ correspondence extends from $d = 2$ to $d = 3$: the ∞-categorical obstruction $\mathrm{HH}^{-2}_{E_1}(C) = 0$ (for smooth proper CY_3 categories) implies that the E_3-lifting of the S^3-framing is contractible, producing an A_∞-structure on the boundary algebra compatible with the chiral bar complex.*

At $d = 2$, the CY-A correspondence was the first proved case: the Fukaya category of a symplectic surface carries an A_∞-structure whose bar construction is the wrapped Floer complex, and the A_∞-formality on the Koszul locus is the E_2-collapse of the bar spectral sequence. At $d = 3$, the ∞-categorical proof resolves the chain-level compatibility of the S^3-framing with the factorization structure on the $\mathbb{C}$-direction by showing that the relevant obstruction group vanishes.

The remaining algebraic prerequisites are explicit. The MC5 boundary analysis is genus-zero and does not include an elliptic corner; the Koszulness characterization still depends on the D-module purity converse; the shadow obstruction tower is constructed at all degrees; the arithmetic invariants are computed; the standard landscape is fully mapped. CY-A at $d = 3$ was the last unresolved input; its proof completes the algebraic foundation.

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Modular Koszul duality for chiral algebras is E_1–E_1 operadic Koszul duality transported into the homotopical modular chiral realm on algebraic curves.

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